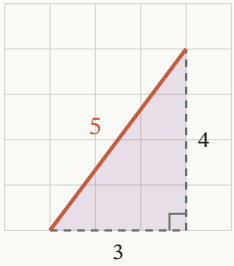
Lines and circles CHEAT SHEET

THE DISTANCE FORMULA



$\text{run}^2 + \text{rise}^2 = \text{distance}^2$

$3^2 + 4^2 = 25 = 5^2$

the order of the points doesn't matter (squares)

$PQ = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$

$OP = \sqrt{(x^2 + y^2)}$ $\quad (3, 4)$: $\sqrt{(9 + 16)} = 5$

$(-2, 3)$ to $(4, -5)$: $\sqrt{(36 + 64)} = 10$

SECTION FORMULA

internal $m : n$: $\left(\frac{mx_2 + nx_1}{m + n}, \frac{my_2 + ny_1}{m + n}\right)$

external $m : n$: $\left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}\right)$

midpoint: $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$

$(-1, 7), (4, -3)$ in $2 : 3 \rightarrow (1, 3)$

CENTROID AND AREA

centroid $G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$

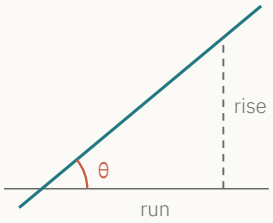
area $= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$

$(1, 1), (5, 2), (2, 5)$: $\frac{1}{2} |-3 + 20 - 2| = 7.5$

area $0 \iff$ the three points are collinear

shoelace: list the corners, repeat the first, cross-multiply

SLOPE



$m = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1} = \tan\theta$

$(-2, 5), (3, -5)$: $m = \frac{-10}{5} = -2$

0°: 0	30°: $1/\sqrt{3} \approx 0.577$
45°: 1	60°: $\sqrt{3} \approx 1.732$
90°: undefined	135°: -1

rising: $m > 0$ · falling: $m < 0$ · flat: 0 · vertical: undefined

PARALLEL AND PERPENDICULAR

parallel: $m_1 = m_2$

perpendicular: $m_1 m_2 = -1$ (neither line vertical)

slopes 2 and $-\frac{1}{2}$: perpendicular

ANGLE BETWEEN TWO LINES

$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$ (acute θ)

slopes 2 and $\frac{1}{3}$: $\left|\frac{\frac{1}{3} - 2}{1 + \frac{2}{3}}\right| = 1 \rightarrow 45°$

$m_1 = m_2$: parallel $\quad 1 + m_1 m_2 = 0$: $90°$

collinear: equal slopes through a shared point

FORMS OF A LINE

axes-parallel	$x = -2$ (vertical) $\quad y = 3$ (horizontal)
point-slope	$y - y_1 = m(x - x_1)$ $(-2, 3)$, $m = -4$: $y = -4x - 5$
two-point	$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1)$ $(1, -1), (3, 5)$: $y = 3x - 4$
slope-intercept	$y = mx + c$ $°F = 1.8\,°C + 32$
intercept	$\frac{x}{a} + \frac{y}{b} = 1$ $x/4 + y/3 = 1$
normal	$x\cos\omega + y\sin\omega = p$ $p = 4$, $\omega = 30°$: $\sqrt{3}x + y = 8$
general	$Ax + By + C = 0$ slope $-A/B$ · y-cut $-C/B$ · x-cut $-C/A$

parallel: $Ax + By + k = 0$ $\quad$ perpendicular: $Bx - Ay + k = 0$

$3x - 4y + 12 = 0$: $y = \frac{3}{4}x + 3$, $\frac{x}{-4} + \frac{y}{3} = 1$

meet: solve together $\quad 2x + y = 7, x - y = -1 \rightarrow (2, 3)$

DISTANCE FROM A LINE

$d = \frac{|Ax_1 + By_1 + C|}{\sqrt{(A^2 + B^2)}}$

parallel lines: $d = \frac{|C_1 - C_2|}{\sqrt{(A^2 + B^2)}}$ (match A, B first)

foot: $(x_1, y_1) - \frac{Ax_1 + By_1 + C}{A^2 + B^2}(A, B)$

$(3, -5)$ and $3x - 4y - 26 = 0$: $\frac{|9 + 20 - 26|}{5} = \frac{3}{5}$

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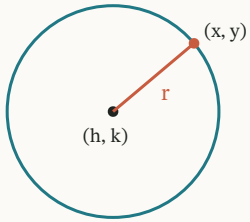
THE CIRCLE



every point at distance r from the centre: the distance formula, held fixed

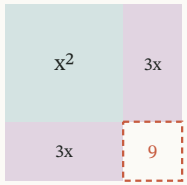
$(x-h)^2+(y-k)^2=r^2$ centre (h, k), radius r

$x^2+y^2+2gx+2fy+c=0$

centre $(-g,-f)$ $r=\sqrt{(g^2+f^2-c)}$

g^2+f^2-c: >0 circle · $=0$ a point · <0 none

COMPLETING THE SQUARE



$x^2+6x=(x+3)^2-9$

the missing corner is $3\times3=9$

$x^2+y^2+8x+10y-8=0 \rightarrow$ centre $(-4,-5)$, $r=7$

a number in front of x^2 and y^2? divide by it first

A CIRCLE FROM POINTS

diameter (x_1, y_1)–(x_2, y_2):

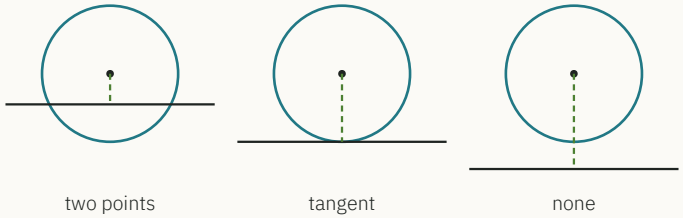
$(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$

three points: put each in, solve for g, f, c

$(5,5), (-2,4), (2,-4) \rightarrow$ centre $(2,1)$, $r=5$

the angle in a semicircle is 90° · bisectors of chords meet at the centre

A LINE AND A CIRCLE



two points | tangent | none

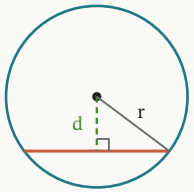
d: the drop from the centre to the line

$d<r$: two points $d=r$: tangent $d>r$: none

or substitute: b^2-4ac >0 · $=0$ · <0

$x^2+y^2=25$, $x+y=7 \rightarrow (3,4)$ and $(4,3)$

LENGTH OF A CHORD



half chord $=\sqrt{(r^2-d^2)}$

the drop from the centre bisects the chord

$x^2+y^2-6x-4y-12=0$, $4x+3y-3=0$:

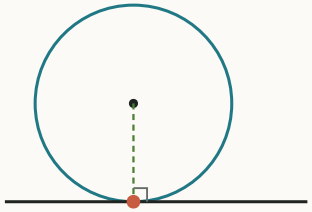
$d=3$, half chord $\sqrt{(25-9)}=4$, chord 8

A POINT AND A CIRCLE

$S_1=x_1^2+y_1^2+2gx_1+2fy_1+c$

$S_1<0$ inside · $=0$ on · >0 outside

TANGENTS



the radius is perpendicular to the tangent

at (x_1, y_1) on $x^2+y^2=r^2$: $xx_1+yy_1=r^2$

$y=mx+c$ touches $x^2+y^2=r^2 \iff c^2=r^2(1+m^2)$

$x^2+y^2=25$ at (3, 4): $3x+4y=25$

FROM A POINT OUTSIDE

tangent length $=\sqrt{S_1}=\sqrt{(PC^2-r^2)}$

$P(7,1)$, $x^2+y^2=25$: $\sqrt{(50-25)}=5$

slopes $4/3$ and $-\frac{3}{4}$: $4x-3y=25$, $3x+4y=25$